OPERATIONAL RUNBOOK v1.0.0

WPT-PRO Operational Playbook & Runbook

System Incident Triage, Diagnostic Workflows & Pipeline State Recovery Protocols

This operational runbook provides System Administrators and Operations teams with standard operating procedures (SOPs), diagnostic workflows, log inspection guidelines, and emergency recovery commands for maintaining **WPT-PRO** and **WPT-PRO Plus** in production.

> **First Responder Emergency Protocol:** If a system appears stuck in a reboot loop or fails to launch post-reboot, do NOT manually delete the entire `C:\WPT\` directory. Execute the *Emergency Pipeline Reset Protocol* detailed on Page 2.

INCIDENT TRIAGE MATRIX

Symptom / Event	Root Cause Analysis	Immediate Action Plan
Pipeline Halts After Reboot #1	Task Scheduler `\WPTPRO_Phase1To2_Handoff` failed to trigger or SYSTEM context lost execution rights.	Inspect Task Scheduler history, verify `C:\WPT\Logs\wpt_execution.log`, or trigger Bridge12 manually via elevated prompt.
Duration Shows < 5 Minutes on Full Run	`FinKPI_v0.ps1` failed to read `Pre_Opt_KPI.json` and fell back to local process start time.	Verify `C:\WPT\DATA\Pre_Opt_KPI.json` existence. Ensure TOM completed successfully without early script exit.
HTML Report Launches Mid-Pass (Weekend Plus)	Report launched before background `wpt-plus` servicing threads returned exit code 0.	Ensure `FinKPI` delay wrapper is active in `wpt-plus.ps1`. Report will auto-refresh upon final completion.
Memory Usage Spikes Post-Optimization	System services re-allocated cache during NGEN pre-compilation pass.	Allow system to idle for 2 minutes; dynamic working set flush will automatically settle available memory to >1.25 GB.

DIAGNOSTIC INCIDENT FLOW



EMERGENCY PIPELINE RESET PROTOCOL

When execution state is compromised or a reboot loop occurs, execute this complete state purge script from an elevated PowerShell prompt to return WPT-PRO to a clean standby state:

```
# Step 1: Terminate active pipeline processes Get-Process -Name "pwsh", "powershell" -ErrorAction SilentlyContinue | Where-Object { $_.CommandLine -like "*WPT*" } | Stop-Process -Force # Step 2: Unregister Scheduled Tasks Unregister-ScheduledTask -TaskName "WPTPRO_Phase1To2_Handoff" -Confirm:$false -ErrorAction SilentlyContinue Unregister-ScheduledTask -TaskName "WPTPRO_Phase2To3_Handoff" -Confirm:$false -ErrorAction SilentlyContinue # Step 3: Clear RunOnce Registry Fallback Keys Remove-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce" -Name "WPTPRO_Transit" -ErrorAction SilentlyContinue # Step 4: Purge Transient State JSON Files Remove-Item -Path "C:\WPT\DATA\Pipeline_State.json" -Force -ErrorAction SilentlyContinue Remove-Item -Path "C:\WPT\DATA\Pre_Opt_KPI.json" -Force -ErrorAction SilentlyContinue Write-Host "[OK] WPT-PRO Pipeline state completely reset." -ForegroundColor Green
```

SCHEDULED TASK MANAGEMENT SOP

Verifying Active Handoff Tasks

To inspect scheduled handoff tasks created during active execution, query the Task Scheduler root folder:

```
Get-ScheduledTask -TaskName "WPTPRO_*" | Select-Object TaskName, State, Actions
```

Manual Task Invocation

If a reboot completed but the handoff task failed to start automatically, manually trigger Phase 2 handoff:

```
Start-ScheduledTask -TaskName "WPTPRO_Phase1To2_Handoff"
```

LOG FILE INSPECTION & PARSING

Unified pipeline events are recorded at `C:\WPT\Logs\wpt_execution.log`. Use the following PowerShell snippet to tail and filter for error events in real time:

```
# Live Log Monitoring (Tail Mode) Get-Content -Path "C:\WPT\Logs\wpt_execution.log" -Wait -Tail 20 # Filter Log for Errors & Warnings Across Recent Runs Get-Content -Path "C:\WPT\Logs\wpt_execution.log" | Select-String -Pattern "ERROR", "WARNING", "CRITICAL"
```

WEEKEND PLUS PASS SPECIAL RUNBOOK

The extended servicing pass (`wpt-plus.ps1`) carries an extended runtime (~2–3 hours) due to deep component store maintenance and NGEN framework re-compilation. Follow these specific operational procedures:

DISM & SFC Lockout Prevention

During deep servicing passes, DISM and SFC acquire exclusive file locks on the Windows Component Store (`C:\Windows\WinSxS`). Ensure no Windows Update background installations or third-party installer agents (e.g., MSIExec) are active prior to launching `wpt-plus.ps1`.

```
# Check for Active Servicing Locks Get-Service -Name "wuauserv", "trustedinstalle
r" | Select-Object Name, Status
```

MANUAL STATE FILE REPAIR SOP

If `Pre_Opt_KPI.json` is accidentally deleted mid-transit, reconstruct the baseline JSON manually to restore Cornerstone duration calculations:

```
{ "Timestamp": "2026-10-09T22:00:00.0000000Z", "CPU_Idle_Percent": 0.0, "Free_RAM_GB":
1.00, "Active_Processes": 80, "Disk_Free_GB": 100.00, "Stage": "Phase1_TOM_Complete" }
```

ROUTINE MAINTENANCE CHECKLIST

- **Monthly Log Trimming:** Archive or truncate `C:\WPT\Logs\wpt_execution.log` if file size exceeds 50 MB.
- **PowerShell 7 Runtime Updates:** Keep binaries in `C:\WPT\pwsh\` updated to current LTS releases.
- **Report Directory Hygiene:** Periodically purge legacy HTML reports in `C:\WPT\Reports\` to preserve storage space.

WPT-PRO Operational Playbook & Runbook — End of Document